SIMATS Engineering • CSE • CSA0804 — Python Programming • 27-07-2026 • Category: Class Work (21.7.26) • Student Submissions Report

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Q1. Sum the given number upto single digit

Score: 5.00 **TC: 1/1 passed**

CODE

```
n = int(input())
while n >= 10:
    total = 0
    while n > 0:
        total += n % 10
        n //= 10
    n = total
print(n)
```

INPUT

```
89789
```

OUTPUT

```
5
```

Q2. Prime Number

Score: 5.00 **TC: 1/1 passed**

CODE

```
def is_prime(n):
    if n <= 1:
        return False
```

```
    if n ≤ 3:
        return True
    if n % 2 == 0 or n % 3 == 0:
        return False
    i = 5
    while i * i ≤ n:
        if n % i == 0 or n % (i + 2) == 0:
            return False
        i += 6
    return True

x = int(input().strip())
n = x + 1
while not is_prime(n):
    n += 1

print(n)
```

INPUT 13

OUTPUT 17

Q3. Swap the first and last digits

Score: 5.00 TC: 1/1 passed

CODE

```
n = input().strip()

if not n:
    print(0)
else:
    first = n[0]
    last = n[-1]
    middle = n[1:-1]
    result = last + middle + first
    print(result)
```

INPUT

```
23546
```

OUTPUT

```
63542
```

Q4. Odd Sequence first and Even Sequence next

Score: 5.00 TC: 1/1 passed

CODE

```
s = input().strip()

odd = []
even = []

for ch in s:
    d = int(ch)
    if d % 2 ≠ 0:
        odd.append(ch)
```

```
    else:
        even.append(ch)

print(''.join(odd + even))
```

INPUT 658134

OUTPUT 513684

Q5. Step Number

Score: 5.00 TC: 2/2 passed

CODE

```
n = input().strip()
digits = list(n)

is_step = True
for i in range(len(digits) - 1):
    if abs(int(digits[i]) - int(digits[i+1])) ≠ 1:
        is_step = False
        break

if is_step:
    print("Yes")
else:
    print("No")
```

INPUT 54567

OUTPUT Yes

Q6. Reverse only the even numbers

Score: 0.00 TC: 0/1 passed

CODE

```
s=input()
even=[c for c in s if int(c)%2==0]
j=len(even)-1
out=[]
for c in s:
    if int(c)%2==0:
        out.append(even[j]); j-=1
    else:
        out.append(c)
print(''.join(out))
```

INPUT 325649

OUTPUT 345629

Q7. Encoding

Score: 5.00 TC: 1/1 passed

CODE

```
s = input().strip()
res = []
n = len(s)
i = 0
while i < n:
    j = i
    while j < n and s[j] == s[i]:
        j += 1
    res.append(s[i] + str(j - i))
```

```
    i = j
print(''.join(res))
```

INPUT 55553322255

OUTPUT 54322352

Q8. Reverse Odd Digit

Score: 5.00 TC: 1/1 passed

CODE

```
s = input().strip()
digits = list(s)

odd = [ch for ch in digits if int(ch) % 2 ≠ 0]
odd.reverse()

j = 0
res = []
for ch in digits:
    if int(ch) % 2 ≠ 0:
        res.append(odd[j])
        j += 1
    else:
        pass

print(''.join(res))
```

INPUT 816253

OUTPUT 351

Q9. Convert every number into the next sequence number.

Score: 5.00 TC: 1/1 passed

CODE

```
s = input().strip()

result = ""
for ch in s:
    d = int(ch)
    result += str((d + 1) % 10)

print(result)
```

INPUT 89781

OUTPUT 90892

Q10. Print the total factor count of a given number

Score: 5.00 TC: 1/1 passed

CODE

```
import math

n = int(input().strip())
count = 0

for i in range(1, int(math.sqrt(n)) + 1):
    if n % i == 0:
        j = n // i
        if i == j:
            count += 1
```

```
            count += 1
        else:
            count += 2

print(count)
```

INPUT 12

OUTPUT 6